

- 1 (a) By considering $(r+1)^2 - r^2$, use the method of differences to prove that

$$\sum_{r=1}^n r = \frac{1}{2}n(n+1). \quad [4]$$

[illegible]

- (b)** Given that $\sum_{r=1}^n (r+a) = n$, find a in terms of n . [3]

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